

# Overview of Respiratory Physiology

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## Learning Objectives (3)

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*After completing this brick, you will be able to:*

- 1 Explain the mechanics of ventilation and gas exchange.
- 2 Outline the significance of mucociliary transport and underlying associated pathology associated with it.
- 3 Describe the structure and function of the alveolar-capillary unit.



## CASE CONNECTION

RM is a 70-year-old retired engineer with emphysema and chronic obstructive pulmonary disease (COPD), and he presents to the emergency department with shortness of breath. “It’s been really bad since the day before yesterday. Even when I sit still, I can’t catch my breath,” he says as he gasps for air. RM has

smoked tobacco since the age of 14; he does not consistently take his COPD medications. On room air, his pulse oximetry is 91%. You decide to give him supplemental oxygen.

How will supplemental oxygen help RM? Consider your answer as you read, and we'll revisit RM at the end of the brick.

[GO TO CONCLUSION](#) ↓

## Introduction

The respiratory system is essential for survival, providing oxygen ( $O_2$ ) for cellular metabolism, and removing the carbon dioxide ( $CO_2$ ) waste product. It also has other important respiratory functions such as phonation (making sounds), generating metabolites (surfactant and ACE), acid base balance ( $CO_2$  and bicarbonate ( $HCO_3^-$ ) buffering), and defense against environmental insults.

# How does oxygen and carbon dioxide get into and out of the lungs (ventilation) (LO1)

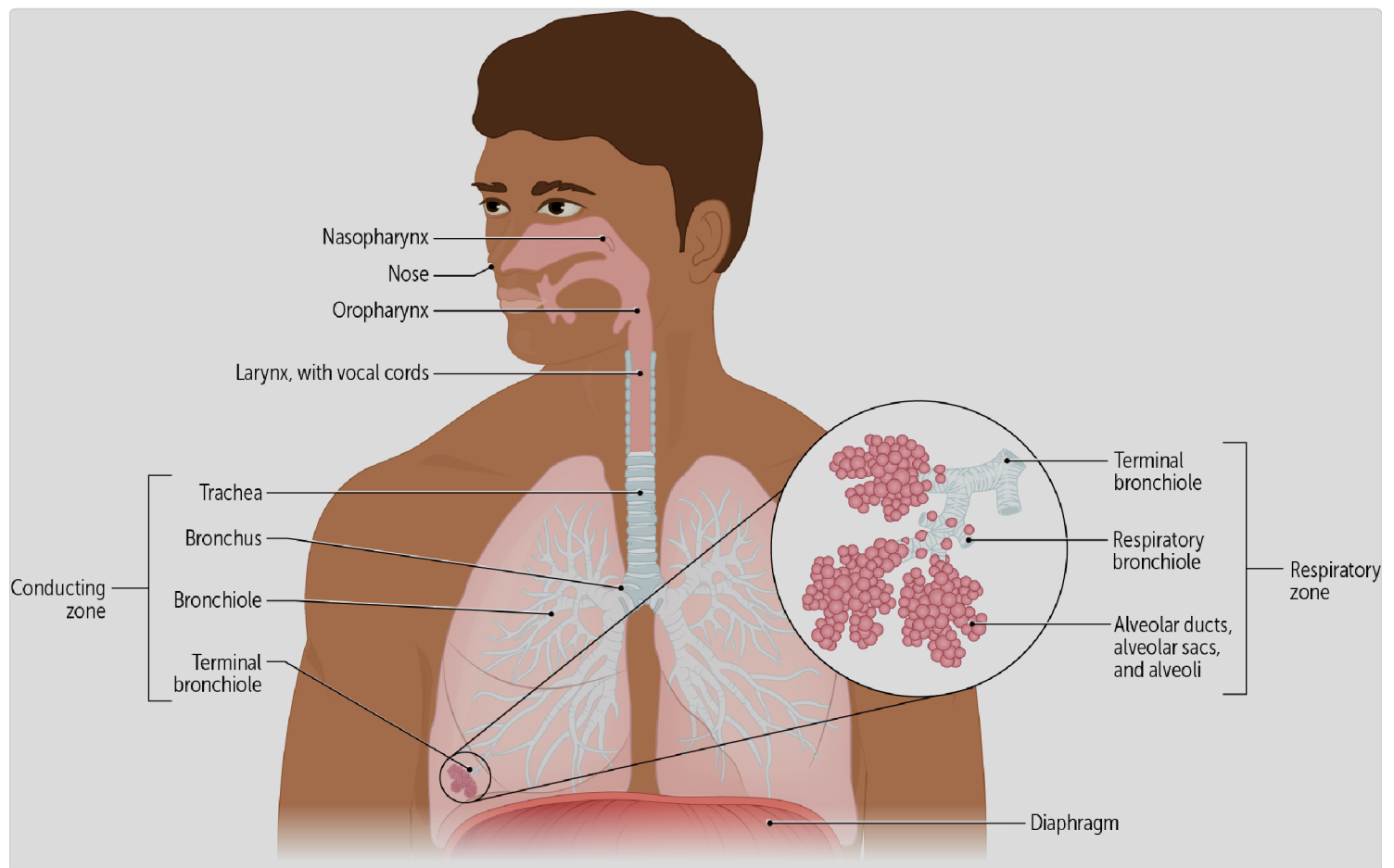


Figure 1 Respiratory Tract

Take a look at [Figure 1](#); air, containing oxygen, is inhaled in the air through the nose and brought into the alveoli. Conversely, the carbon dioxide generated by cellular metabolism is exhaled in the reverse direction. *Let's engage in a reflective exercise to explore aspects of inhalation and exhalation.*

**Inhalation:** Close your eyes and focus on your breathing. You can place a hand on your chest if you wish to feel the movements better. Feel your abdomen expand with the diaphragm contraction and flattening. Next, as your rib cage moves up, the thorax expands first, increasing its volume. These

volume and pressure changes occur before air movement. Next you feel the air entering your nose and travelling deep into your lungs. This is because a pressure gradient was created, and the air flowed from high to low pressure. Now hold your breath at the end of inhalation. Do you feel that this breath holding takes effort?

**Exhalation:** Now let the air exit your lungs. You will notice that the chest volume changes slightly before the airflow changes. Your rib cage moves down, and the air exits due to the elastic recoil of your lungs. These changes result in a pressure gradient in the opposite direction, with a higher pressure in the lungs this time and lower pressure at your mouth. Hold your breath at the end of exhalation; does it take effort? Was exhalation longer than inhalation?

This ventilation of moving air into and out of the lungs will be covered in more detail in a later brick. The take-home messages for this section are that 1. inhalation takes effort, while exhalation is passive and slower at rest, and that 2. changes in volume then pressure gradient drives the movement of air into and out of the lungs.

(relative to atmospheric pressure) in the alveoli in the lungs.

The pressure gradient between the mouth and the negative pressure

## Immune Function of the Upper Respiratory Tract (LO2)

A lot of air moves in and out of your lungs every day, along with dust and pathogens. The upper respiratory tract helps defend the body from these environmental pathogens. For example, during passage through the nasal cavity, mucus captures inhaled particles, ciliary motion promotes their removal, and the air is conditioned via humidification. These "unwanted" particles trapped early on in the nasal passages are either sneezed, blown out, or swallowed. Particles that escape such filtering in the nasal passages are trapped in mucus in the trachea or bronchioles and brought upwards by the mucociliary escalator to be expectorated, coughed out, or swallowed.

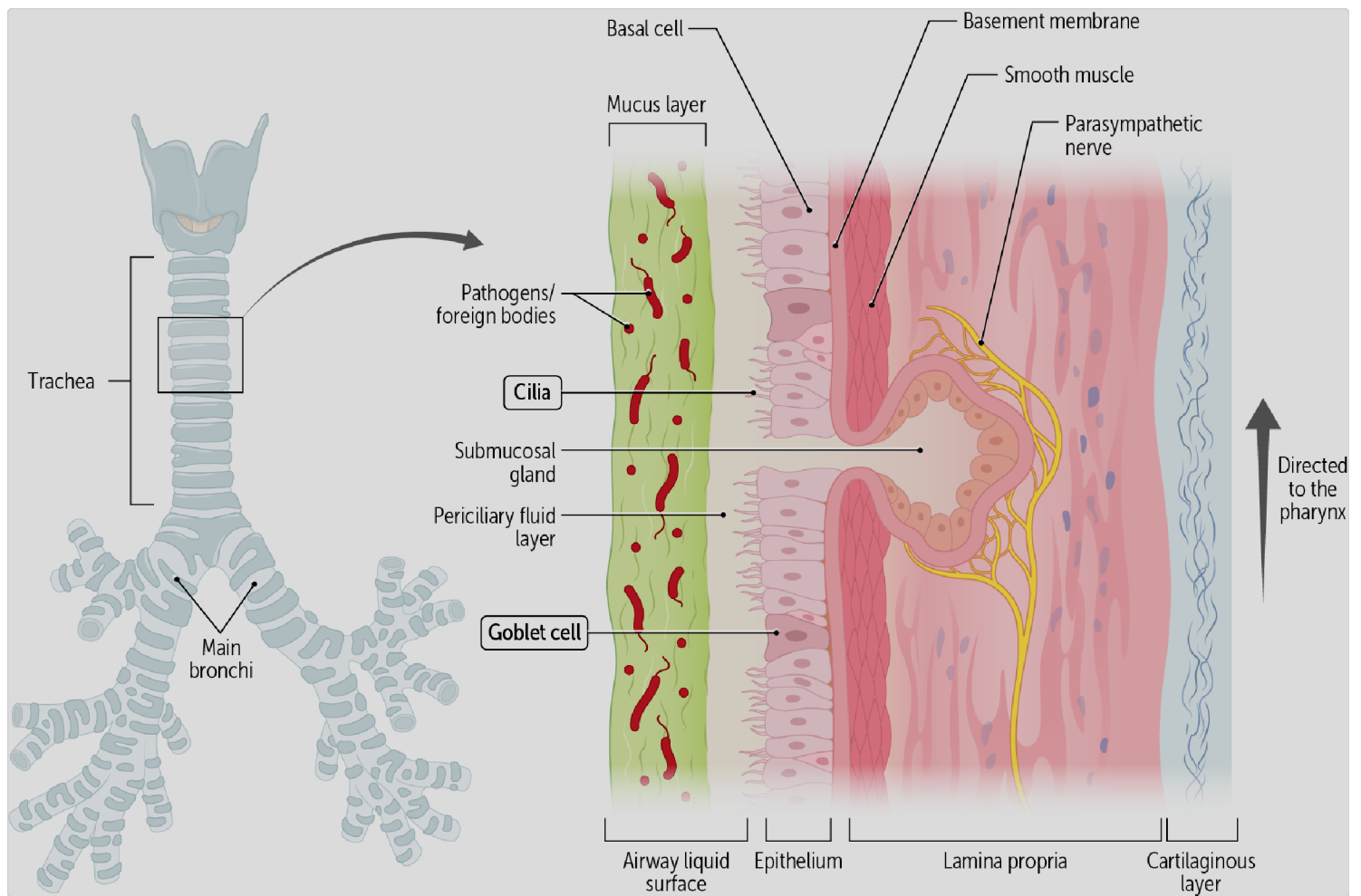
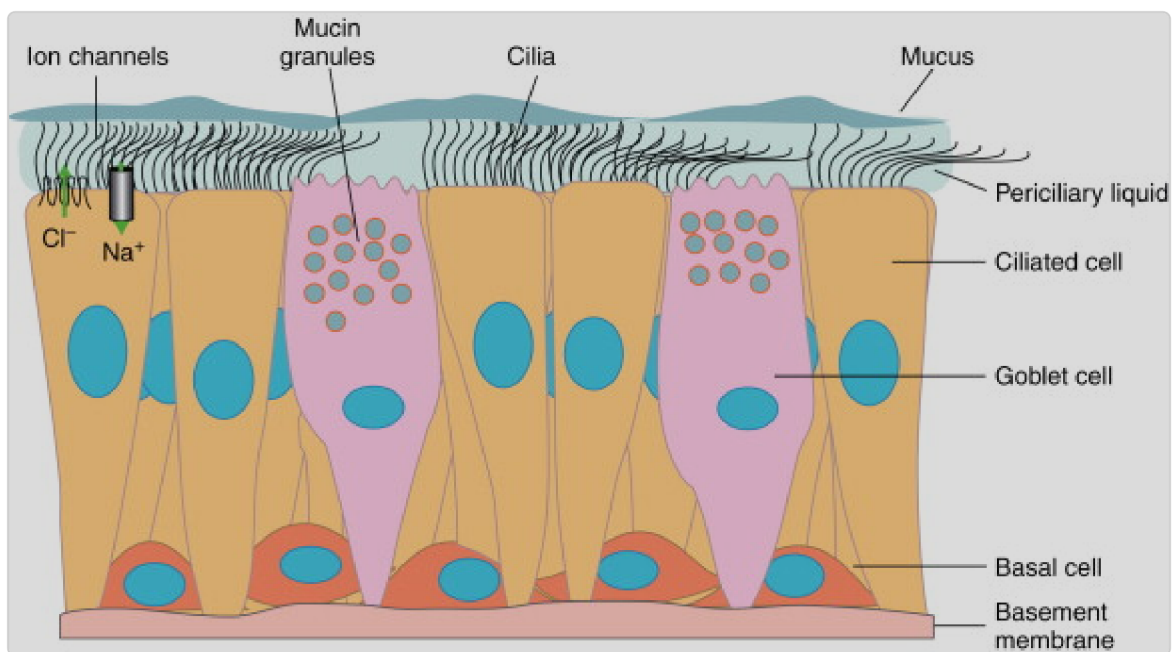


Figure 2a. Histological appearance of the mucociliary escalator



**Figure 2b. How cilia and mucus work to in the mucociliary escalator. The cilia beat and move mucus upwards toward the mouth.**

**Mucociliary escalator** is present in the conducting airways and involves a coordinated activity of cilia and mucus (**Figure 2a**). The mucociliary escalator has several key microscopic components, including goblet cells, submucosal glands (mucus secreting), and cilia (motile appendages). Ciliated epithelial cells and mucus-secreting goblet cells line the conducting airways, extending from the posterior nasopharynx and trachea through the bronchi and bronchioles, with goblet cells progressively decreasing and largely absent at the level of the terminal bronchioles. The cilia beat rhythmically and move mucus and trapped unwanted particles upward in a mucus "stream" towards the mouth, away from the lungs (**Figure 2b**), much like an escalator bringing people up to the next floor. The mucus carried upwards is usually swallowed. Greater amounts of mucus accumulation (due to either increased production or reduced clearance) can trigger cough reflexes as the body tries to clear the airways. Excessive or abnormal mucus accumulation becomes pathological in diseases such as asthma (mucus hypersecretion), cystic fibrosis (dehydrated, viscous secretions with impaired clearance), and bronchiectasis (chronic infection-driven mucus hyperproduction and retention).

### CLINICAL CORRELATION

**Smoking** cigarettes can reduce the efficacy of the mucociliary escalator. Cigarettes contain toxins that reduce ciliary beat frequency, disrupt the coordinated beating pattern, and reduce ciliary length. This can lead to

increased infections because the lungs are less able to protect themselves from pathogens.

**Alveolar macrophages:** Beyond the terminal bronchioles, airway epithelium lacks cilia, and particle clearance depends primarily on **alveolar macrophages**. Thus, pulmonary defense shifts from mucocilliary transport in the conducting airways to macrophage-mediated clearance in the gas-exchange regions. These long-lived sentinel cells reside on the alveolar surface, where they phagocytose inhaled particles, microorganisms, and cellular debris. In addition to debris clearance, alveolar macrophages play a central role in initiating and sustaining pulmonary immune responses through cytokine release and antigen presentation. Dysfunction or overload of these macrophages contributes to several high-yield pathologies. For example, inhaled mineral dusts such as silica impair macrophage function and promote chronic inflammation and fibrosis, while carbon dust accumulation leads to characteristic pigment-laden macrophages in coal workers' pneumoconiosis.

Below the functions of the immune/protective cells in the airways are elucidated:

- **Goblet cells** (in the trachea to bronchioles) produce mucus that covers the respiratory epithelium and traps particles or microorganisms (**Figure 2a**). Parasympathetic nerves stimulate mucous secretion.
- **Submucosal glands** (in the trachea and bronchi) also produce mucus, as well as proteins that have antimicrobial action (**Figure 2a**).
- **Cilia** are hair-like projections found on the surface of the respiratory epithelial cells. They are motile and beat rhythmically upward to push the mucus and trapped particles or microorganisms out of the

respiratory tract and into the throat, where they are swallowed or coughed up (Figure 2b).

- **Club cells**, abundant in the terminal bronchioles, where mucus-secreting goblet cells decrease in number, contribute to airway protection by secreting protective proteins and metabolizing inhaled toxins. These cells also function as progenitor cells that support bronchiolar epithelial repair following injury (not shown).
- **Immune cells**: The respiratory epithelium is patrolled by neutrophils, macrophages, and other immune cells, which, along with the secretion of IgA and antimicrobial peptides, help neutralize pathogens (not shown).



#### CLINICAL CORRELATION

**Mucus hypersecretion and accumulation** are seen in conditions such as asthma, bronchitis, bronchiolitis, pneumonia, and cystic fibrosis. It can obstruct the airways, leading to airflow limitation and atelectasis (collapse of alveoli). Stagnant mucus also promotes bacterial growth, increasing the risk of infection and worsening the underlying disease.

they are swallowed or coughed up.

mucus and moves them away from the lungs toward the throat where

The mucociliary escalator traps particles and microorganisms in

## Diffusion across the alveolar–capillary unit or alveolar barrier (LO3)

Once in the lungs, the gases diffuse across the gas-liquid barrier. This barrier, while very thin, consists of several layers. Describing the layers from alveoli to blood, these layers consist of the following ([Figure 2](#)):

- **Fluid and surfactant layer**—a thin layer of fluid on the alveolar surface,
- **Alveolar epithelium**- a single cell layer of thin squamous epithelium consisting of type 1 pneumocytes and cuboidal type II pneumocytes (not shown).
- **Epithelial and capillary basement membranes**—there are two basement membranes, with the interstitial space (connective tissue such as elastin) in between them
- **Capillary epithelium**—a thin layer of squamous epithelium making up the continuous capillaries (without fenestrae or "holes") as in other types of capillaries (fenestrated or sinusoidal/discontinuous).

Damage to the layers of the alveolar barrier can affect both getting air into and out of the lungs and gas diffusion.

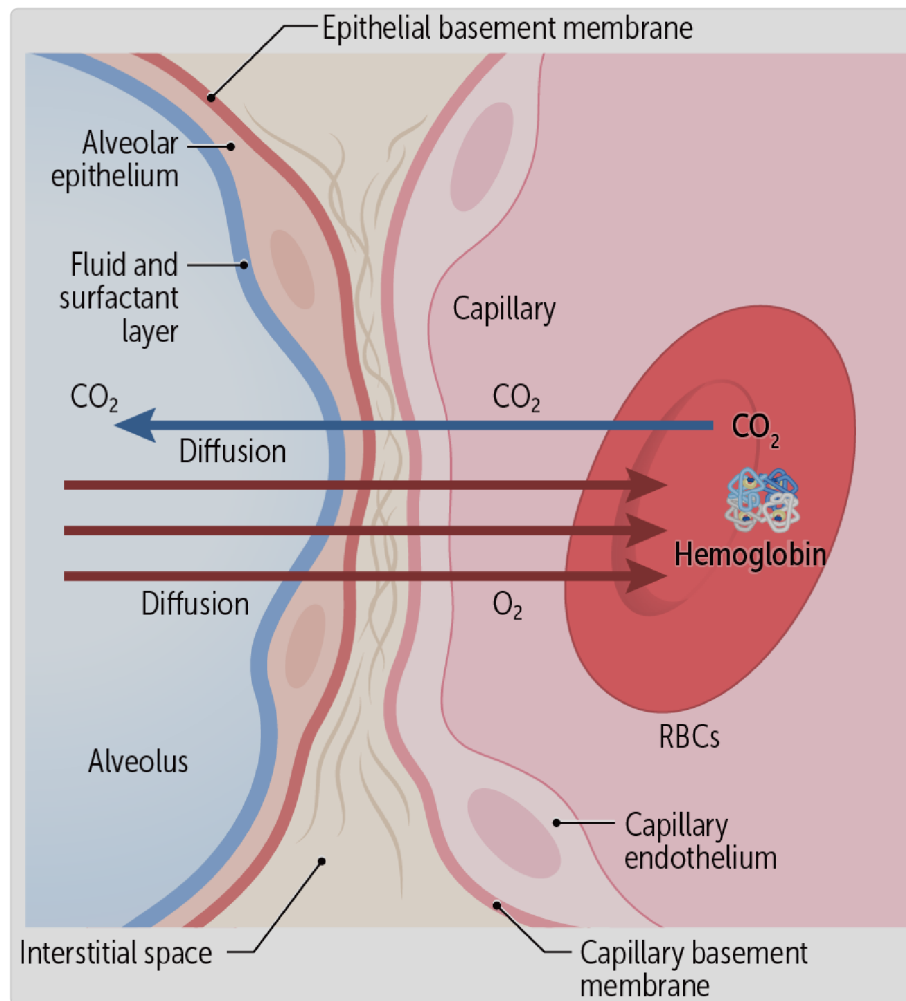


Figure 2. Alveolar diffusion barrier

So, what factors affect gas diffusion across this barrier?

**Fick's law of diffusion:**  $\dot{V}_{\text{gas}} = (A \times D \times (P_1 - P_2)) / T$

Where:  $\dot{V}_{\text{gas}}$  = rate of gas diffusion **A** = surface area of the membrane **D** = diffusion coefficient of the gas **P<sub>1</sub> - P<sub>2</sub>** = difference in partial pressure of the gas across the membrane **T** = thickness of the membrane. (You do not have to

*memorize this equation but Fick's law factors play a role in most of the respiratory processes and diseases.)*

**Fick's law of diffusion** outlines factors that affect gas movement across the alveolar barrier. Fick's law indicates that diffusion is proportional to **increases** in the concentration or pressure gradient ( $P_1 - P_2$ ) of the gas ( $O_2$  or  $CO_2$ ), surface area, and diffusibility of the gas, and is **reduced** by greater membrane thickness ([Figure 3](#)).

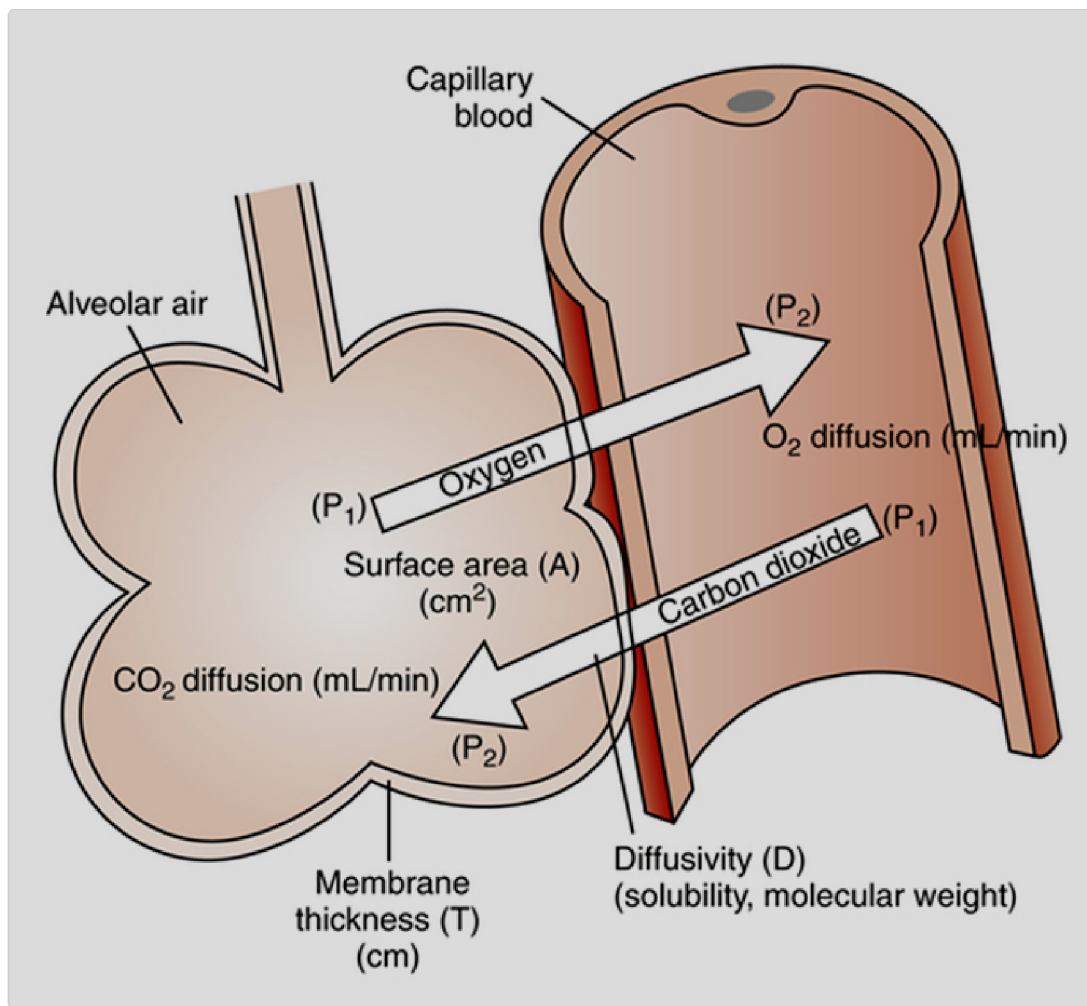


Figure 3. Fick's factors that affect diffusion

## CLINICAL CORRELATION

### Application of Fick's laws to disease:

In **pulmonary fibrosis**, fibrotic *thickening* of the interstitium and loss of alveolar capillaries slows oxygen diffusion. As a result, patients develop hypoxemia (low blood oxygen levels), which often worsens during exertion when blood is moving more quickly through capillaries (decreased capillary transit time), such as what happens during exercise. This leads to exercise intolerance and desaturation (lowered blood O<sub>2</sub> saturation), frequently requiring supplemental oxygen to increase the alveolar-to-blood diffusion gradient.

In **emphysema**, *destruction of alveoli and their capillaries* greatly reduces the surface area available for gas exchange. Loss of alveoli and elastin also impairs recoil, decreasing the alveolar pressures that help force air out. This leads to air trapping in the lungs in this obstructive disease. The resulting enlarged, poorly perfused air spaces further limit oxygen diffusion, especially during exertion. Smoking is the leading cause of emphysema, and many patients require supplemental oxygen to enhance oxygenation by increasing the diffusion gradient.

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It decreases due to thickened alveolar walls.

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The pressure gradient between alveoli and pulmonary capillaries.

## Diffusion of Gas across the Alveolus to Artery – Pressures and Concentrations

There are different ways to measure a gas, and in turn the efficacy of gas movement through the alveolar barrier. Of the Fick's factors, the **partial pressure gradient** is the primary driving force for gas movement. Partial of dissolved gases is written with a large P, and is the gas that is dissolved in liquid. Partial pressure reflects the kinetic energy of freely moving gas molecules, the force that propels molecules from one compartment to another. Gases diffuse from regions of higher partial pressure to regions of lower partial pressure. At the alveolus, oxygen diffuses into capillary blood because alveolar  $P_{A}O_2$  (~100 mmHg) exceeds the dissolved  $PO_2$  of incoming deoxygenated arteriolar blood (~40 mmHg). Carbon dioxide diffuses in the opposite direction because  $CO_2$  dissolved withing blood at the capillary  $PCO_2$  (~46 mmHg) exceeds that of alveolar  $P_{A}CO_2$  (~40 mmHg).

A critical principle follows from this: **only dissolved gas molecules contribute to partial pressure**. Oxygen bound to hemoglobin is sequestered — it does not exert partial pressure and therefore does not participate in or influence diffusion. This is actually what makes hemoglobin so effective: by binding dissolved  $O_2$  as it enters the capillary, hemoglobin continuously removes free oxygen from plasma, keeping dissolved  $PO_2$  low and preserving

the diffusion gradient. Without this binding, plasma PO<sub>2</sub> would equilibrate with alveolar PO<sub>2</sub> almost immediately, and diffusion would stop long before adequate oxygen had been transferred.

Table 1: Diffusion across the alveolar membrane is affected by the below Fick's law factors

| Factor                                                       | Role in Ficks Law     | How it Influences Diffusion                                                                        | Physiologic / Clinical Example                                                                             |
|--------------------------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Partial Pressure Gradient (P <sub>1</sub> – P <sub>2</sub> ) | Gradients drive Flow  | Diffusion increases as the pressure difference between alveolar gas and capillary blood increases. | Supplemental O <sub>2</sub> increases the alveolar–capillary gradient, enhancing O <sub>2</sub> diffusion. |
| Surface Area (A)                                             | Proportional factor   | Diffusion increases with greater available membrane area.                                          | Emphysema reduces alveolar surface area → decreased DLCO.                                                  |
| Diffusion Coefficient (D)                                    | Gas-specific constant | Determines how easily a gas diffuses based on solubility and molecular weight.                     | CO <sub>2</sub> diffuses more readily than O <sub>2</sub> due to higher solubility.                        |
| Membrane Thickness (T)                                       | Inverse factor        | Diffusion <b>decreases</b> as the diffusion barrier becomes thicker.                               | Pulmonary fibrosis increases membrane thickness → impaired gas transfer.                                   |

 **CASE CONNECTION**

**BACK TO INTRODUCTION** 

Thinking back to RM’s case — During normal breathing, diaphragm contraction draws oxygen into the lungs and expels carbon dioxide during exhalation. In RM’s COPD, damaged alveoli and increased airway resistance disrupt these ventilatory mechanics and reduce the effective surface area for gas exchange,

both key factors in Fick's law. As a result, oxygen diffusion into the blood is impaired.

Administering supplemental oxygen raises the alveolar oxygen concentration and steepens the pressure gradient for diffusion across the alveolar membrane (another Fick's law factor), enhancing oxygen transfer to the blood. This improves hemoglobin saturation and helps restore tissue oxygenation despite his compromised lung function.

## Summary

- The respiratory system is essential for oxygen supply, carbon dioxide removal, phonation, metabolite generation, acid-base balance, and defense against environmental insults.
- Ventilation, or breathing, involves the movement of gases (primarily oxygen and carbon dioxide) driven by pressure gradients between the atmosphere and the alveoli during inhalation and exhalation.
- Inhalation, which requires effort due to diaphragm contraction and thoracic expansion, and passive exhalation is driven by lung elastic recoil.
- The mucociliary escalator in the upper respiratory tract traps and removes particles and pathogens, preventing them from reaching the lungs.
- Oxygen from the air diffuses from the alveoli into the surrounding alveolar capillaries, and carbon dioxide produced by tissue metabolism, diffuses from the blood into the alveoli.
- Gas diffusion across the alveolar barrier is governed by Fick's law, which considers pressure gradient, diffusion constant, surface area, and membrane thickness.

- Application of Fick's law: In diseases like pulmonary fibrosis and emphysema, gas diffusion is impaired due to changes in the alveolar barrier such as increased membrane thickness or reduced surface area.

## Review Questions

1. During inspiration, what mechanism drives air into the lungs?

- A. The contraction of abdominal muscles
- B. Positive pressure in the atmosphere compared to the lungs
- C. Negative pressure within the alveoli
- D. Relaxation of the diaphragm
- E. Increased airway resistance

Explanation (requires correct answer)

2. In pulmonary fibrosis, what direct effect does the thickened alveolar membrane have on gas diffusion?

- A. Decreases the rate of oxygen diffusion
- B. Increases the surface area for gas exchange
- C. Decreases the partial pressure difference between alveoli and blood

- D. Reduces CO<sub>2</sub> diffusion but not O<sub>2</sub> diffusion
- E. Enhances oxygen uptake

**Explanation** (requires correct answer)

3. Under normal conditions, which of the following cellular structures is responsible for moving trapped particles and mucus upward toward the pharynx?

- A. Cilia
- B. Goblet cells
- C. Smooth muscle
- D. Submucosal glands

**Explanation** (requires correct answer)

**How would you rate this Brick?**



## References

- Bennett, J. (2020). ClinicalKey. Respiration: control of ventilation

### Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01** Explain the mechanics of ventilation and gas exchange.
- 02** Outline the significance of mucociliary transport and underlying associated pathology associated with it.
- 03** Describe the structure and function of the alveolar-capillary unit.

**Objective Confidence: 0/3**